# UNEMPLOYMENT RATE

during 2019 - 2021

PRESENTATION BY BA PHAM

MAY 2026

# INTRODUCTION

None of us will forget the COVID-19 pandemic. The world was in a panic, many people died, and the global economy stopped. Between 2019 and 2021, the world and the U.S. economy changed a lot.

# MY HYPOTHESIS

Between 2019 and 2021, the unemployment rate in tourism-dependent states, such as Nevada and Hawaii, fluctuated significantly more than that in tech-driven states, such as California and Washington

# DATASET

## Collecting

Collecting from usda.org
Link:
https://www.ers.usda.gov/data-products/atlas-of-rural-and-small-town-america

## Cleaning

Focusing on
Tourism-dependent states:
- Nevada
- Hawaii

Tech-driven states
- California
- Washington

# COMMAND LINE

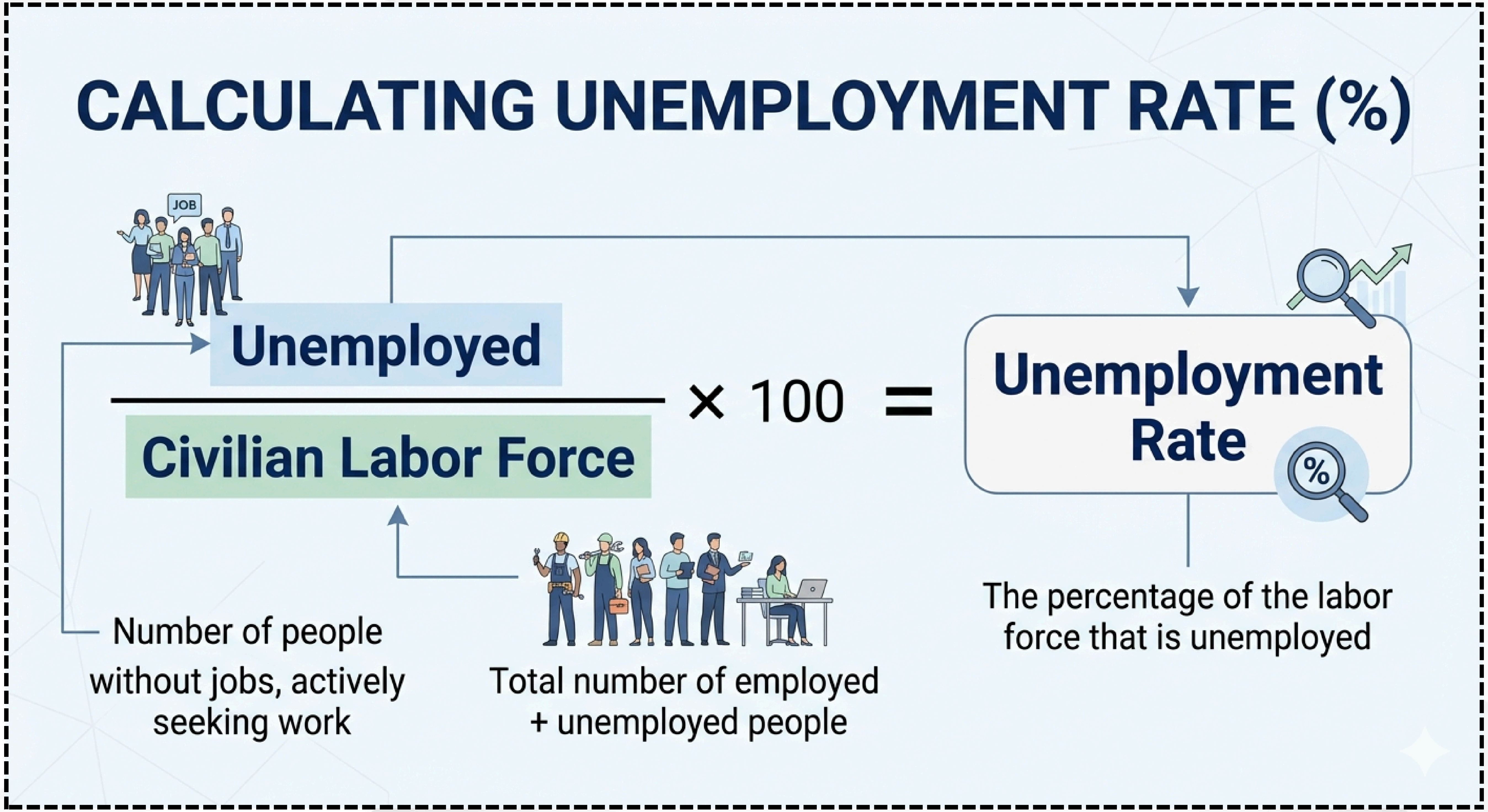
awk -F, '{print $2, $3, $58, $59, $41, $43, $51, $53}' Jobs_data.csv > unemploy_

```
hoang@Harry-Bapham MINGW64 ~/Documents/HW1_DataIntensiveClass/HW1_data_intensive_cl
$ awk -F, '{print $2, $3, $58, $59, $41, $43, $51, $53}' Jobs_data.csv > unemploy_l

hoang@Harry-Bapham MINGW64 ~/Documents/HW1_DataIntensiveClass/HW1_data_intensive_cl
$ head -5 unemploy_labor.txt
State County NumUnemployed2019 NumCivLaborforce2019 NumUnemployed2020 NumCivLaborFo
US United States 6009990 163815890 13023736 161207103 8658650 161758337
AL Alabama 72024 2271892 147160 2268309 77272 2246993
AL Autauga 764 26684 1420 26405 742 26341
AL Baldwin 2877 98921 6159 98910 2946 99427
```

| Field | Column number |
|---|---|
| State | 2 |
| County | 3 |
| NumUnemployed2019 | 58 |
| NumUnemployed2020 | 41 |
| NumUnemployed2021 | 51 |
| NumCivLaborforce2019 | 59 |
| NumCivLaborforce2020 | **43** |
| NumCivLaborforce2021 | 53 |

# COMMAND LINE



# COMMAND LINE

```
awk 'NR==1    {print}
> $1 == "HI" && $2 == "Hawaii"    {print}
> $1 == "NV" && $2 == "Nevada"    {print}
> $1 == "CA" && $2 == "California"    {print}
> $1 == "WA" && $2 == "Washington"    {print}
> ' Unemploy_labor.txt > 4_states.txt
```

# COMMAND LINE

| State | UnemployRate2019 | UnemployRate2020 | UnemployRate2021 |
|---|---|---|---|
| CA | 4.11% | 10.22% | 7.30% |
| HI | 2.45% | 12.00% | 5.72% |
| ~~HI~~ | ~~3.13%~~ | ~~11.67%~~ | ~~5.51%~~ |
| NV | 4.02% | 13.52% | 7.23% |
| WA | 4.30% | 8.47% | 5.23% |

# SPARK

**Select necessary columns**

```
df = df.select(df['State'], df['County'], df['NumUnemployed2019'],df['NumUnemployed2020'],df['NumUnemployed2021'],df['NumCivLabor
df.show()
```

```
+-----+-------------+-----------------+-----------------+-----------------+--------------------+--------------------+----------
----------+
|State|       County|NumUnemployed2019|NumUnemployed2020|NumUnemployed2021|NumCivLaborforce2019|NumCivLaborforce2020|NumCivLabo
rforce2021|
+-----+-------------+-----------------+-----------------+-----------------+--------------------+--------------------+----------
----------+
|   US|United States|          6009990|         13023736|          8658650|           163815890|           161207103|
161758337|
|   AL|      Alabama|            72024|           147160|            77272|             2271892|             2268309|
2246993|
|   AL|      Autauga|              764|             1420|              742|               26684|               26405|
26341|
|   AL|      Baldwin|             2877|             6159|             2946|               98921|               98910|
99427|
```

SPARK

# Filter rows that consist of such states as Nevada, Hawaii, California, and Washington

**Filter necessary rows**

```
from pyspark.sql import functions as F
df_states = df.filter(
    ((F.col('State') == 'NV') & (F.col('County') == 'Nevada')) |
    ((F.col('State') == 'HI') & (F.col('County') == 'Hawaii')) |
    ((F.col('State') == 'CA') & (F.col('County') == 'California')) |
    ((F.col('State') == 'WA') & (F.col('County') == 'Washington'))
)
df_states.show()
```

```
+-----+----------+-----------------+-----------------+-----------------+--------------------+---
-------+
|State|    County|NumUnemployed2019|NumUnemployed2020|NumUnemployed2021|NumCivLaborforce2019|Num
rce2021|
+-----+----------+-----------------+-----------------+-----------------+--------------------+---
-------+
|   CA|California|           796806|          1934450|          1381250|            19409413|
8923194|
|   HI|    Hawaii|            16776|            79512|            38226|              684690|
568413|
```

# SPARK

## Create the UnemploymentRate column

**Create UnemploymentRate column = NumUnemployed column / NumCivlLaborforce column**

```
from pyspark.sql import functions as F
df_div = df_states.withColumns({
    'UnemployedRate2019': F.round(F.col('NumUnemployed2019') / F.col('NumCivLaborforce2019') * 100,2),
    'UnemployedRate2020': F.round(F.col('NumUnemployed2020') / F.col('NumCivLaborforce2020') * 100,2),
    'UnemployedRate2021': F.round(F.col('NumUnemployed2021') / F.col('NumCivLaborforce2021') * 100,2)
                            })
df_div.show()
```

```
+-----+----------+-----------------+-----------------+-----------------+--------------------+----------
-------+------------------+------------------+------------------+
|State|    County|NumUnemployed2019|NumUnemployed2020|NumUnemployed2021|NumCivLaborforce2019|NumCivLabor
rce2021|UnemployedRate2019|UnemployedRate2020|UnemployedRate2021|
+-----+----------+-----------------+-----------------+-----------------+--------------------+----------
-------+------------------+------------------+------------------+
|   CA|California|           796806|          1934450|          1381250|            19409413|
8923194|              4.11|             10.22|               7.3|
|   HI|    Hawaii|            16776|            79512|            38226|              684690|
668413|              2.45|              12.0|              5.72|
|   NV|    Nevada|            61995|           203208|           108822|             1541772|
```

# SPARK

| State | County | UnemployedRate2019 | UnemployedRate2020 | UnemployedRate2021 |
|---|---|---|---|---|
| CA | California | 4.11 | 10.22 | 7.3 |
| HI | Hawaii | 2.45 | 12.0 | 5.72 |
| NV | Nevada | 4.02 | 13.52 | 7.23 |
| WA | Washington | 4.3 | 8.47 | 5.23 |

Final Table

# COMPARASION

| State | UnemployRate2019 | UnemployRate2020 | UnemployRate2021 |
|---|---|---|---|
| CA | 4.11% | 10.22% | 7.30% |
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| ~~HI~~ | ~~3.13%~~ | ~~11.67%~~ | ~~5.51%~~ |
| NV | 4.02% | 13.52% | 7.23% |
| WA | 4.30% | 8.47% | 5.23% |

Homework 1

Homework 2

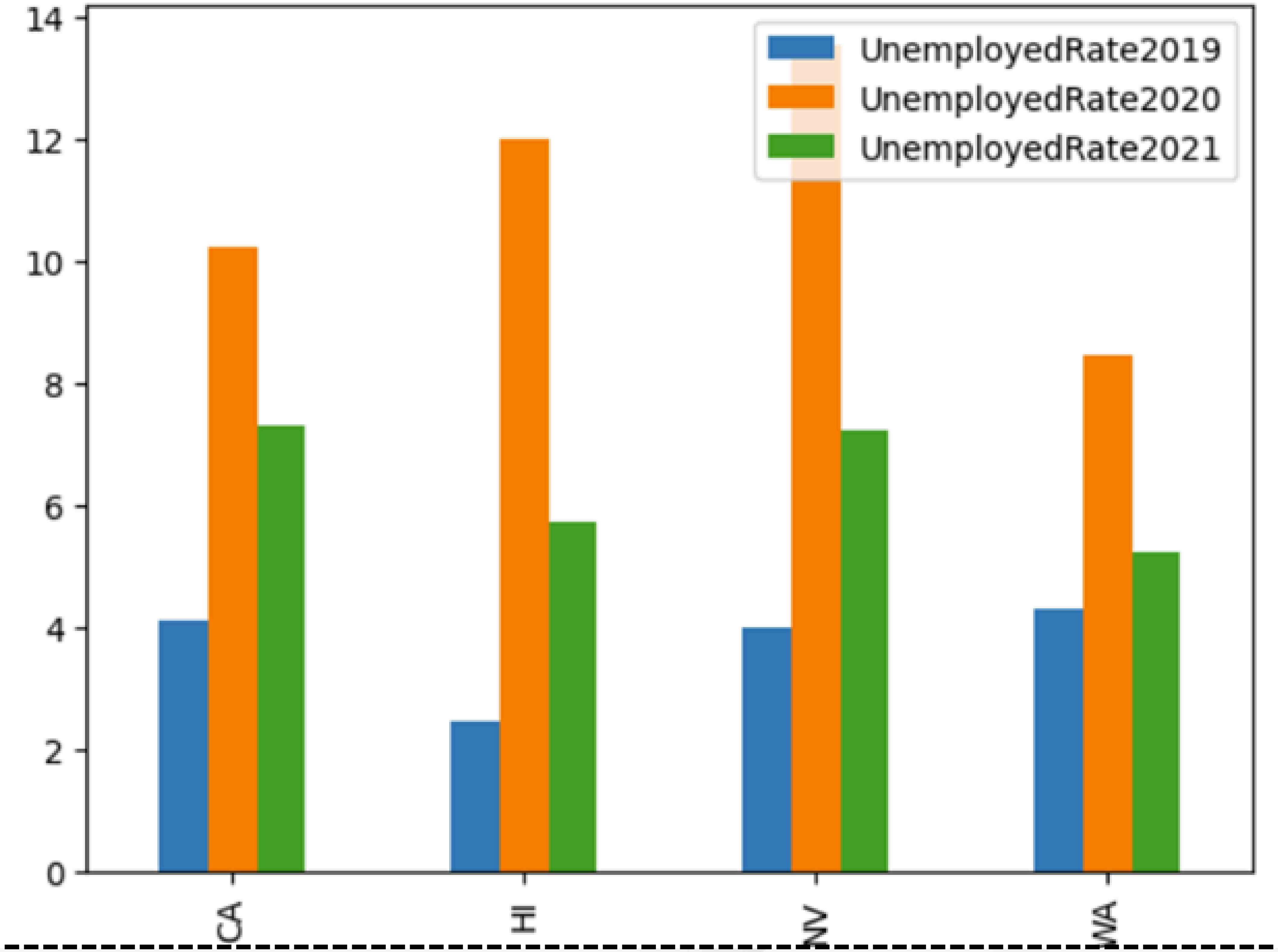
| State | County | UnemployedRate2019 | UnemployedRate2020 | UnemployedRate2021 |
|---|---|---|---|---|
| CA | California | 4.11 | 10.22 | 7.3 |
| HI | Hawaii | 2.45 | 12.0 | 5.72 |
| NV | Nevada | 4.02 | 13.52 | 7.23 |
| WA | Washington | 4.3 | 8.47 | 5.23 |

# VISUALIZATION

| State | 2019 | 2020 | 2021 |
|---|---|---|---|
| HI | 2,45 | 12,00 | 5,72 |
| NV | 4,02 | 13,52 | 7,23 |
| CA | 4,11 | 10,22 | 7,30 |
| WA | 4,30 | 8,47 | 5,23 |

```
df_rate.toPandas().plot.bar(x ='State')
```

<Axes: xlabel='State'>



# CONCLUSION

Between 2019 and 2021, the unemployment rate in tourism-dependent states, such as Nevada and Hawaii, fluctuated significantly more than those in tech-driven states, such as California and Washington

# THANKYOU